# CASE STUDY 8

# MANUFACTURING COST AND PROFITABILITY ANALYSIS

Python Programming – Activity-Based Learning

## 1. Problem Statement

Write a Python program to calculate the total manufacturing cost of a product. The program takes material cost, labour cost, wastage cost, machine cost, number of products and selling price per product. It calculates total revenue and profit, then checks whether the required profit target is achieved.

## 2. Objectives

• To understand input and arithmetic operations in Python.
• To calculate total production cost and revenue.
• To find profit or loss.
• To use if-else to check a profit target.

## 3. Assumptions

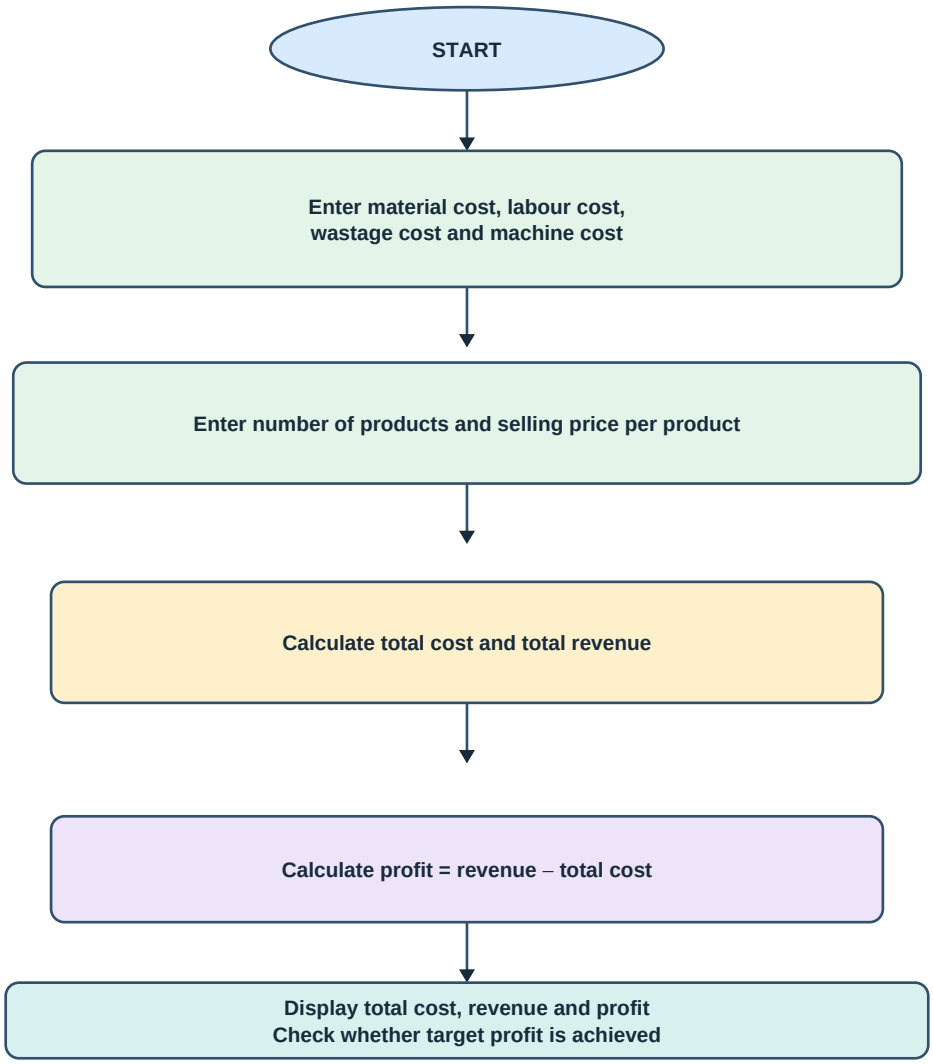
• All costs and selling price are entered in rupees.
• Number of products must be greater than zero.
• Costs and selling price cannot be negative.
• Total cost = material + labour + wastage + machine cost.
• Revenue = number of products × selling price per product.
• Required profit target is ■5,000 for this example.

## 4. Algorithm

1. Start the program.
2. Input material, labour, wastage and machine costs.
3. Input number of products and selling price per product.
4. Check that inputs are valid.
5. Calculate total cost by adding all costs.
6. Calculate revenue by multiplying quantity and selling price.
7. Calculate profit = revenue – total cost.
8. Compare profit with the target of ■5,000.
9. Display the results and stop.

## 5. Flowchart

Colour flowchart showing the manufacturing cost and profit calculation:


START
Enter material cost, labour cost,
wastage cost and machine cost
Enter number of products and selling price per product
Calculate total cost and total revenue
Calculate profit = revenue – total cost
If profit < target: target not achieved
Display total cost, revenue and profit
Check whether target profit is achieved

## 6. Pseudocode

```
START
INPUT material_cost
INPUT labour_cost
INPUT wastage_cost
INPUT machine_cost
INPUT quantity
INPUT selling_price

IF any cost or price is negative OR quantity <= 0
    PRINT "Invalid input"
ELSE
    total_cost = material_cost + labour_cost
                 + wastage_cost + machine_cost
    revenue = quantity * selling_price
    profit = revenue - total_cost

    PRINT total_cost
    PRINT revenue
    PRINT profit

    IF profit >= 5000
        PRINT "Profit target achieved"
    ELSE
        PRINT "Profit target not achieved"
    END IF
END IF
STOP
```

## 7. Python Source Code

```
material = float(input("Enter material cost: "))
labour = float(input("Enter labour cost: "))
wastage = float(input("Enter wastage cost: "))
machine = float(input("Enter machine cost: "))
quantity = int(input("Enter number of products: "))
price = float(input("Enter selling price per product: "))

if material < 0 or labour < 0 or wastage < 0 or machine < 0 or quantity <= 0 or price < 0:
    print("Invalid input")
else:
    total_cost = material + labour + wastage + machine
    revenue = quantity * price
    profit = revenue - total_cost

    print("Total Manufacturing Cost =", total_cost)
    print("Total Revenue =", revenue)
    print("Profit =", profit)

    if profit >= 5000:
        print("Profit target achieved")
    else:
        print("Profit target not achieved")
```

## 8. Test Cases

| Test | Total Cost | Quantity | Price/Product | Profit | Expected Result |
|---|---|---|---|---|---|
| 1 | 10000 | 100 | 200 | 10000 | Target achieved |
| 2 | 15000 | 100 | 180 | 3000 | Target not achieved |
| 3 | 12000 | 100 | 100 | -2000 | Target not achieved (loss) |
| 4 | — | 0 | — | — | Invalid input |

## 9. Output Screenshots

Sample Run 1 – Profit target achieved:

```
Enter material cost: 4000
Enter labour cost: 2000
Enter wastage cost: 1000
```

```
Enter machine cost: 3000
Enter number of products: 100
Enter selling price per product: 200
Total Manufacturing Cost = 10000.0
Total Revenue = 20000.0
Profit = 10000.0
Profit target achieved
```

Sample Run 2 – Profit target not achieved:

```
Enter material cost: 6000
Enter labour cost: 3000
Enter wastage cost: 2000
Enter machine cost: 4000
Enter number of products: 100
Enter selling price per product: 180
Total Manufacturing Cost = 15000.0
Total Revenue = 18000.0
Profit = 3000.0
Profit target not achieved
```

## 10. Conclusion

Through this case study, I learned how to calculate manufacturing cost, revenue and profit using Python. I also learned to use if-else conditions to check whether a given profit target is achieved. This program is a simple example of cost and profit calculation in a manufacturing unit.

End of Case Study 8